Detecting Cyanobacterial Threats using Machine Learning & Low-Cost Water Quality Sensors

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Abstract

Around the world, cyanobacteria blooms, also known as blue-green algae, release harmful cyanotoxins that contaminate water supplies: disrupting aquatic ecosystems and endangering public health. However, current methods of detection require extensive testing and are too expensive for many at-risk communities. This project aimed to improve the accessibility of cyanobacterial threat detection, using machine learning and low cost water quality sensors.

5 machine learning algorithms (Logistic Regression, Random Forest, Gradient Boosting, XGBoost, Multi-Layer Perceptron) were trained on inexpensive water quality sensor data (turbidity, conductivity, temperature, pH) obtained from around 1,000 lakes surveyed in the EPA National Lakes Assessment. Model performance on the dataset was then optimized using hyperparameter tuning, 5 fold cross validation, and 6 engineered features. Additionally, individual algorithms were ensembled into voting and stacking classifiers.

The best performing model (Stacking Ensemble of Logistic Regression, XGBoost, and Neural Network) achieved a ROC-AUC score of 0.7723 and a 87.9% recall using a modified threshold of 0.35 to prioritize reducing false negative predictions. It was then deployed on a Raspberry Pi integrated with low-cost sensors and mounted on a stationary floating platform. The system effectively outputted cyanobacterial threat predictions during field testing on a local water body; all detections and sensor information were relayed through a real time dashboard during monitoring.

This work demonstrates the feasibility of accessible cyanobacterial monitoring systems, providing a foundation for future AI-driven public health solutions in under-resourced communities.

Introduction

Cyanobacteria, also known as blue-green algae, is a type of microscopic organism frequently found in freshwater and aquatic environments. It is a naturally found primitive bacteria which helps to sustain the world's oxygen atmosphere. Cyanobacteria releases oxygen through photosynthesis; notably there was no life on land prior to the release of oxygen into the atmosphere by cyanobacteria. Additionally, they provide nitrogen from the atmosphere to facilitate the spread of energy in aquatic food webs (epa.gov). However, cyanobacteria can multiply rapidly in warmer months. This excess growth of cyanobacteria, commonly called "blooms", can potentially cause adverse effects on aquatic ecosystems and human health (epa.gov).

Cyanobacteria blooms, under the right conditions (warmer temperature and nutrient abundance) can release harmful cyanotoxins, which pose serious risks to aquatic environments. They can cause serious illness and death to the ecosystem's wildlife, disrupting its vital processes. Additionally, it can threaten humans through bloom contamination of drinking water, or any other bloom exposure through swallowing or inhalation. Symptoms of this exposure can include headaches, cramps, diarrhea, nausea and vomiting, numbness, dizziness, fever, and even more serious conditions based on an individual's vulnerability (oregon.gov).

Cyanobacteria bloom threats are commonly noticed in freshwater environments including wetlands, lakes, reservoirs, streams and rivers. They are becoming more frequent and severe due to climate change and nutrient pollution. Its growth can contaminate drinking water affecting reliant communities, as well as restrict recreational and economic use of aquatic environments (nih.gov).

Accurately detecting cyanobacterial threats requires extensive sampling and testing, beyond visible indicators. According to the EPA, physical and chemical contributing factors of blooms include, nutrient overloading, light availability, water temperature, alteration of water flow, stratification/vertical mixing, pH changes, and trace metals. Additional factors include the ratio of nitrogen to phosphorus, organic nutrient supply, temperature, and light attenuation, which play important and interacting roles in bloom composition (epa.gov). Testing for all of these parameters requires specialized laboratory equipment and trained personnel, making comprehensive bloom monitoring prohibitively expensive for many communities.

Recent research has explored machine learning approaches on metrics such as those above to enhance bloom prediction capabilities. A 2023 study by Lin et al demonstrated effective algal bloom prediction using data-driven machine learning models in a well-monitored lake, showing that when comprehensive datasets including lake nutrient observations, hydrodynamic data, and chlorophyll concentrations were available across multiple predictive workflows, machine learning could reliably predict bloom events. Additionally, novel research has focused not just on ground based detection, but additionally on satellite imagery based monitoring. Hill et al. (2019) developed HABNet, a machine learning system for harmful algal bloom detection using remote sensing data, achieving strong detection performance (91% classification accuracy). However this required 12 distinct data modalities including NASA satellite-derived chlorophyll concentrations and sea surface temperature data, making it dependent on expensive space-based infrastructure. A review by Wang & Qin, 2025 further highlights the use of optical remote sensing in effective harmful algae bloom detection, demonstrating its potential for large scale monitoring across wide geographic areas.

However, all three of these otherwise promising approaches share a critical limitation: they depend on either expensive satellite infrastructure or extensive multi-parameter data collection requiring specialized equipment and trained personnel, making them largely inaccessible to under-resourced communities despite being the areas that often need monitoring the most. A study by Maneyar et al 2025 emphasizes accessible bloom detection due to the fact that despite advancements in cyanobacteria bloom detection modern solutions are heavily biased towards well funded areas (nih.gov). This leaves vast areas of the world underrepresented and at continued risk for toxic blooms.

Thus, this research focuses on bridging the gap between affordability and effectiveness in cyanobacterial threat detection. Instead of extensive ground based metrics which can prove costly, we are focusing on four basic water quality standards: temperature, ph, turbidity, and conductivity. These four measurements were selected because they reflect known environmental conditions that can promote cyanobacterial growth: warmer temperatures can accelerate bloom formation, elevated pH can accompany active photosynthesis during blooms, higher turbidity can indicate bloom density, and conductivity can reflect the nutrient-rich conditions that fuel rapid cyanobacterial multiplication.We are using modern machine learning algorithms and techniques to enhance the effectiveness of predicting cyanobacterial bloom threats just on low cost sensor data. Thus this project hypothesizes that Machine learning models trained on low-cost water quality sensor data can achieve ROC-AUC > 0.75 in predicting cyanobacterial presence, demonstrating feasibility for deployment on affordable intelligent machine platforms for real-time aquatic monitoring.

Methodology

1. Dataset and Preprocessing

Water quality data was obtained from the EPA's National Lakes Assessment 2017, which surveyed lakes across the United States. Three datasets were merged: phytoplankton count data containing cyanobacteria biovolume measurements by algal group, water profile data containing sensor measurements at surface depth (less than or equal to 1.0m) including temperature, pH, and conductivity, and water chemistry data containing turbidity measurements. Cyanobacteria percentage was calculated as cyanobacteria biovolume divided by total phytoplankton biovolume multiplied by 100 for each lake. Samples with cyanobacteria percentage greater than 10% were labeled as "bloom" (1), otherwise "no bloom" (0). The 10% cyanobacteria biovolume threshold was selected based on EPA guidelines for harmful algal bloom monitoring, with modification to enable early bloom detection before blooms reach critical advisory levels. Surface measurements at depth less than or equal to 1.0m were averaged for each lake, all three datasets were merged by lake UID, and incomplete records with missing values were removed. After preprocessing, the final dataset contained 941 lake samples, each with the 4 sensor measurements as features and a binary bloom label as the target variable. The class distribution was 453 lakes with blooms (48.1%) and 488 lakes without blooms (51.9%). The dataset was split into 80% training (752 lakes) and 20% testing (189 lakes) using stratified sampling to maintain class balance.

Table Dataset Statistics

A)

Metric	Value
Total Lakes	941
Training Set	80%
Testing Set	20%
Lakes with Threat Present	488
Lakes without No Threat	453
Balance Ratio	0.93

Summary of the EPA National Lakes Assessment 2017 dataset composition and class distribution after preprocessing

B)

Feature	Min	Max	Mean +- SD	Unit
Temperature	8.29	32.33	23.621+-3.967	°C
pH	3.11	10.6	7.782+-1.027	-
Turbidity	0.12	720	11.93+-46.202	NTU
Conductivity	1.7	35975	640.89+-2208.43	µs/cm

Statistical summary of four water quality sensor measurements (surface water, depth less than or equal to 1.0 m) used as model input features.

2. *Machine Learning Models and Optimization*

Five machine learning algorithms were selected from the scikit-learn library to represent diverse methodologies: Logistic Regression as a linear baseline, Random Forest as a tree-based ensemble method, Gradient Boosting as a sequential boosting algorithm, XGBoost as an optimized gradient boosting implementation, and Multi-Layer Perceptron as a neural network. Each model was first trained on the dataset using default parameters to establish baseline performance. Hyperparameter tuning was then performed to maximize the effectiveness of each algorithm. Two tuning techniques were used: GridSearchCV, which is an exhaustive search testing all possible combinations, was used for Logistic Regression and MLP, while RandomizedSearchCV, which randomly samples 100 combinations from the parameter space, was used for Random Forest, Gradient Boosting, and XGBoost. All models employed 5-fold cross-validation, where the training data is split into 5 equal parts, the model trains on 4 parts and tests on the remaining 1, repeating 5 times so each part serves as the test set once, with final performance being the average across all 5 iterations. Each algorithm was tuned to maximize ROC-AUC score, which measures how well the model distinguishes between bloom and no-bloom classes, with scores closer to 1.0 indicating better performance. ROC-AUC was selected as the primary evaluation metric because it is robust to class imbalance, considers all possible decision thresholds, and is widely used in water quality prediction and environmental monitoring. The default prediction threshold of 0.5 was evaluated and adjusted to 0.35 for final predictions, because in water quality monitoring, missing an actual bloom (false negative) is more dangerous than issuing an unnecessary warning (false positive). This lower threshold increases recall while accepting slightly lower precision, and ROC-AUC scores remain unchanged as they are threshold-independent.

3. *Feature Engineering*

Feature engineering is the process of creating new variables from existing data to help machine learning models detect patterns more easily. While raw sensor readings provide valuable information, combining and transforming them based on scientific knowledge of cyanobacteria biology can reveal relationships that models might otherwise miss. Six features were engineered from the original four sensor measurements based on previous research on bloom formation. WARM_WATER flags when temperature is greater than 20°C, as cyanobacteria blooms grow best in warm water between 20-30°C. HIGH_PH flags when pH is greater than 8.5, because when cyanobacteria photosynthesize they consume CO_2 which raises the water's pH to 8.5-10. HIGH_CONDUCTIVITY flags when conductivity is in the top 25% of all samples, as high conductivity indicates nutrient-rich conditions that blooms need to grow. PH_DEVIATION measures distance from neutral pH of 7.0, showing how far pH is from normal with larger values indicating unusual water chemistry. TEMP_PH_INTERACTION multiplies temperature by pH, combining warm water and high pH which together indicate active blooms more strongly than either measurement alone. TEMP_COND_INTERACTION multiplies temperature by conductivity, combining warm water with high nutrients to capture ideal bloom conditions. All five machine learning models were retrained on the expanded 10-feature dataset using the same hyperparameter tuning procedures, and performance was compared between models trained on original features versus engineered features to measure the impact of feature engineering.

4. *Ensemble Methods*

Two types of ensemble methods were tested to combine predictions from multiple models to potentially achieve better performance than any single model. A Stacking Ensemble Classifier uses predictions from multiple base models as inputs to a meta-learner model. Logistic Regression was used as the meta-learner to combine base model predictions, and 5-fold cross-validation was used during stacking training to prevent overfitting. Various combinations of base models were tested. A Soft Voting Ensemble Classifier averages the predicted probabilities from multiple models, and various combinations of base models were also tested for this method. Ensemble performance was compared to individual models to determine if combining models provided meaningful improvement.

5. *Feasibility Deployment*

To demonstrate the real world feasibility of this research, the best performing model was deployed on a Raspberry Pi connected to four low-cost sensors: a temperature sensor, pH probe, conductivity probe, and turbidity sensor. The floating platform was built using PVC pipes as pontoons and a waterproof junction box to secure the Raspberry Pi and sensor housing. It was first tested in a controlled indoor environment to verify buoyancy and sensor connectivity before being deployed on a local water body for real-world field testing. The total cost of the entire system was around $300, demonstrating that communities can monitor water quality threats inexpensively.

Stationary floating platform with Raspberry Pi and four water quality sensors (temperature, pH, conductivity, turbidity) for real-time bloom detection *(left)*. Real-time web dashboard displaying live sensor readings and bloom risk assessment, from platform. The system shows current water quality metrics and predicts bloom probability using the trained machine learning model. Dashboard created using Claude Code with Python and Flask *(right)*.

Results

Baseline Model Performance

Baseline performance refers to how each model performs using its default parameters, before any optimization. This establishes a starting point to measure the impact of later improvements. Each of the five algorithms was trained and evaluated on the original four sensor features (temperature, pH, conductivity, and turbidity). Table 2 shows the performance metrics for each model. Gradient Boosting achieved the highest baseline ROC-AUC (0.7072), while XGBoost performed lowest (0.6134).

Model	Accuracy	Recall	F1-Score	ROC_AUC
Gradient Boosting	0.7090	0.6593	0.6857	0.7072
Logistic Regression	0.6667	0.7143	0.6736	0.6684
Random Forest	0.6614	0.6044	0.6322	0.6593
Neural Network	0.6243	0.7582	0.6603	0.6291
XGBoost	0.6138	0.6044	0.6011	0.6134

Table 2 Baseline Model Performance. Performance metrics for five machine learning algorithms using default parameters on original features (4 sensors).

Hyperparameter Tuning

Hyperparameters are settings that control how a machine learning model learns and are set before training begins. Hyperparameter tuning is the process of testing different combinations of these settings to find the version that produces the best performance. GridSearchCV, which exhaustively tests all possible combinations, was used for Logistic Regression and MLP. RandomizedSearchCV, which randomly samples 100 combinations from the parameter space, was used for the tree-based models. All models used 5-fold cross-validation, where the training data is split into 5 equal parts, the model trains on 4 and tests on 1, repeating until each part has served as the test set, with final performance being the average across all 5 iterations. Table 3 shows model performance after hyperparameter optimization on the original features. Overall, hyperparameter tuning improved performance on average for the 5 models by around 8%. The best performing individual model after tuning was XGBoost (ROC-AUC: 0.7695).

Model	Accuracy	Recall	F1-Score	ROC_AUC
Random Forest	0.6349	0.7912	0.6761	0.7318
Gradient Boosting	0.6720	0.8571	0.7156	0.6786
XGBoost	0.6720	0.8681	0.7182	0.7695
Logistic Regression	0.6402	0.8681	0.6991	0.7604
Neural Network	0.6667	0.7473	0.6834	0.7446

Table 3 Hypertuned Model Performance. Performance metrics after hyperparameter optimization on original features, showing improvement from baseline.

Feature Engineering

Feature engineering is the process of creating new variables from existing data to help machine learning models detect patterns more easily. While raw sensor readings provide valuable information, combining and transforming them based on scientific knowledge of cyanobacteria biology can reveal relationships that models might otherwise miss. Six features were engineered from the original four measurements: WARM_WATER, HIGH_PH, HIGH_CONDUCTIVITY, PH_DEVIATION, TEMP_PH_INTERACTION, and TEMP_COND_INTERACTION. All five models were retrained on the expanded 10-feature dataset. Table 4 shows model performance after retraining on the engineered features. Feature engineering added a marginal average improvement of around 2% in the ROC-AUC metric across all models. Figure 3 shows the distribution of each sensor measurement between bloom and no-bloom lakes — the overlapping distributions indicate that some lakes share similar sensor readings regardless of bloom status, which creates ambiguity that limits model accuracy. Figure 4 shows feature importance rankings from the Random Forest Classifier, with turbidity (0.172) and pH (0.155) ranking highest among all features. 3 out of the 6 engineered features also had an importance score greater than 0.10. Figure 6 shows the Pearson correlation matrix across all features, with the strongest positive correlations with bloom presence observed for turbidity, pH, and the engineered interaction terms, confirming their predictive value.

Model	Accuracy	Recall	F1-Score	ROC_AUC
Logistic Regression	0.6455	0.8571	0.6996	0.7589
Random Forest	0.6614	0.8791	0.7143	0.7487
Gradient Boosting	0.6667	0.8681	0.7149	0.7608

Neural Network	0.6878	0.7692	0.7035	0.7587
XGBoost	0.6402	0.8681	0.6991	0.7649

Table 4 Hypertuned Model with Feature Engineering. Final performance metrics after hyperparameter optimization on engineered dataset (10 features: 4 original + 6 engineered).

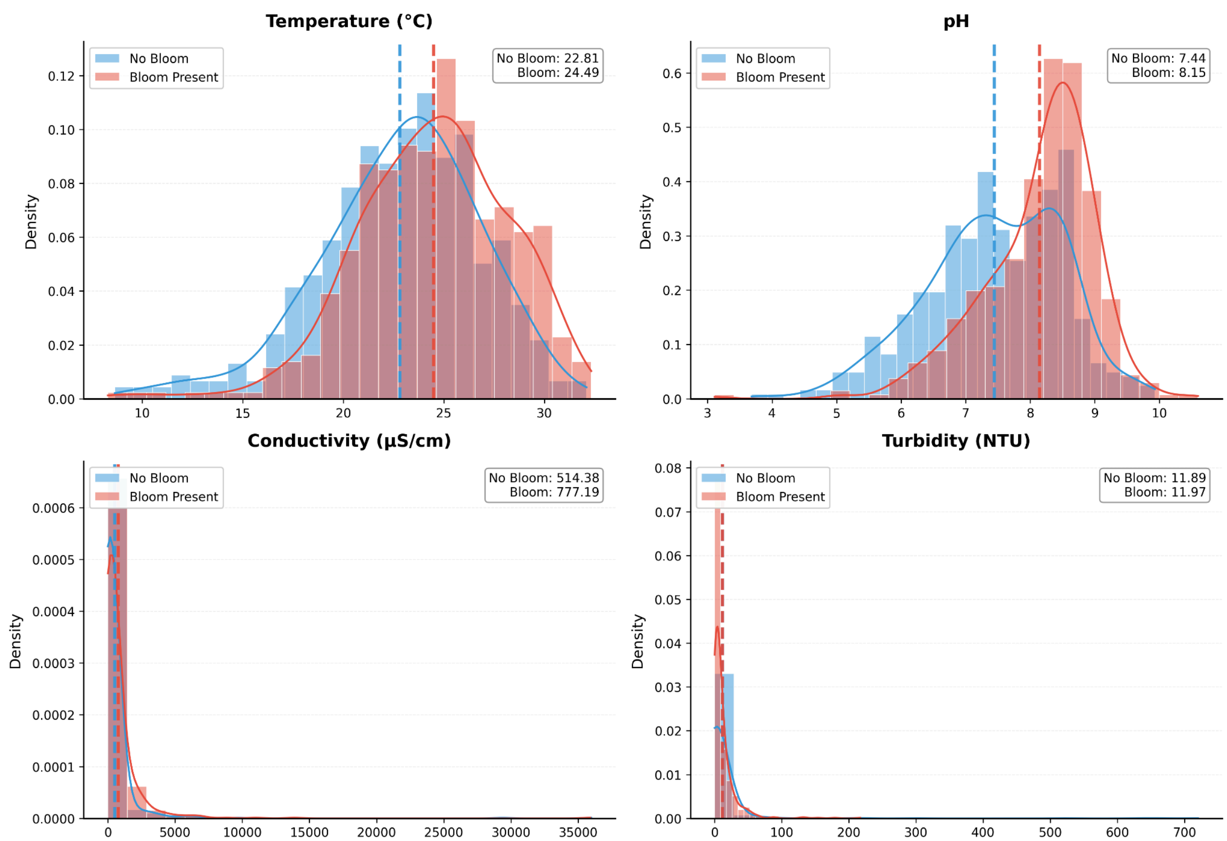

Figure 3 Feature Distributions: Bloom v. No Bloom Conditions**.** Comparison of water quality measurements between lakes with blooms (red) and without blooms (blue). Each histogram shows how the four sensor readings differ between the two groups, with dashed lines marking average values. Overlapping distributions indicate that some lakes share similar sensor readings regardless of bloom status, creating ambiguity that limits model accuracy. Figure created using

Python 3.12 with matplotlib and pandas.

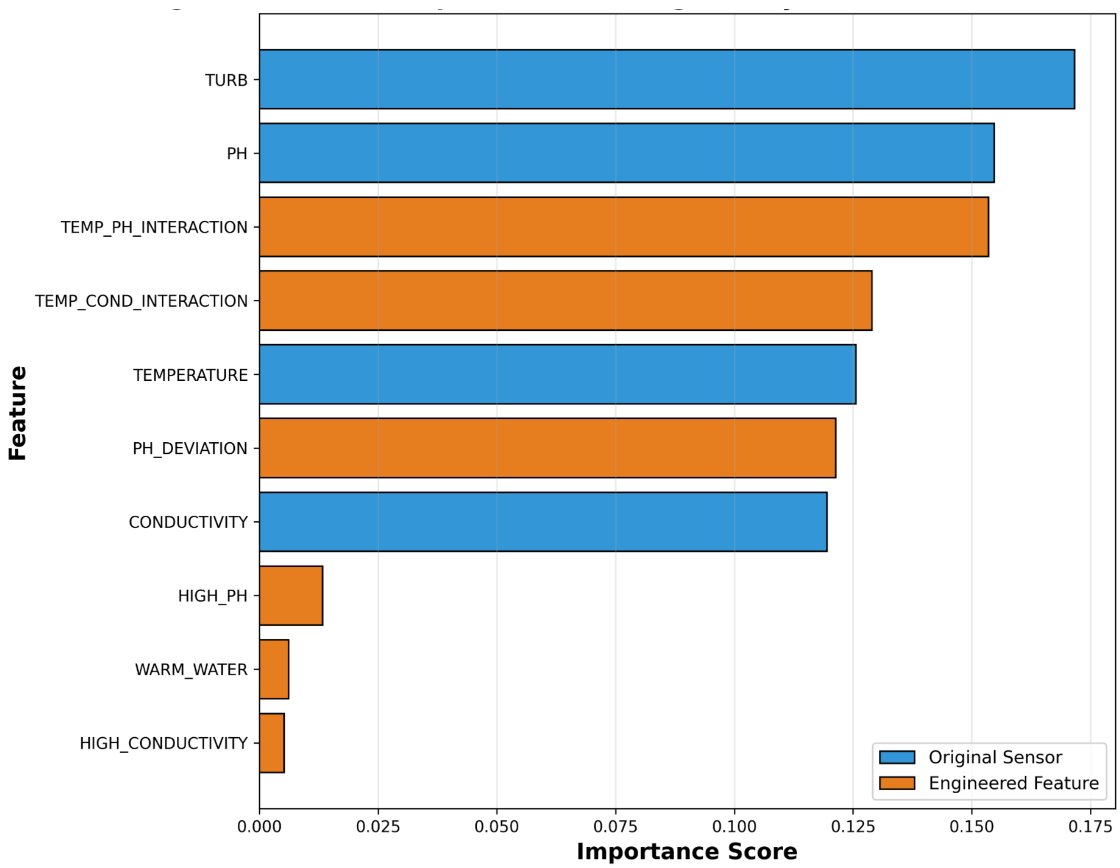

Figure 4 Feature importance from Random Forest Classifier model. Blue bars show original sensors, orange bars show engineered features. Turbidity (0.172) and pH (0.155) ranked highest. 3 out of the 6 engineered features also had an importance greater than 0.10. Figure created using Python 3.12 with matplotlib and scikit-learn.

Ensemble Methods

Ensemble methods combine predictions from multiple models to potentially achieve better performance than any single model. Two types were tested. A Stacking Ensemble Classifier uses predictions from multiple base models as inputs to a meta-learner — in this case Logistic Regression — which learns how to best combine them. A Soft Voting Ensemble Classifier averages the predicted probabilities from multiple models directly. Various combinations of base models were tested for both methods. Table 5 shows the top 10 ensemble methods ranked by ROC-AUC. The best performing ensemble was a Stacking Classifier of Logistic Regression, XGBoost, and Neural Network trained on the feature engineered dataset, achieving a ROC-AUC of 0.7723 and 87.9% recall. Ensembling only showed marginal improvement over the best individual model, ranging between +0.0035 and +0.0074 in ROC-AUC. Figures 1 and 2 show the ROC-AUC scores and ROC curves for each individual model and the best ensemble. All 5 individual models outperformed random guessing (AUC = 0.50).

Method	Models	ROC-AUC	Improvement Over Best Individual
Stacking	lr + mlp + xgb	0.7723	0.0074
Voting	lr + xgb	0.7721	0.0072
Stacking	mlp + xgb	0.7712	0.0063
Voting	lr + gb + mlp + xgb	0.7711	0.0062
Voting	lr + rf + mlp + xgb	0.7711	0.0062
Stacking	lr + xgb	0.7709	0.006
Voting	mlp + xgb	0.7709	0.006
Voting	lr + rf + gb + mlp + xgb	0.7701	0.0052
Stacking	lr + gb + mlp + xgb	0.7698	0.0049
Stacking	lr + gb	0.7684	0.0035

Table 5 Ensemble Method performance. Top 10 ensemble methods ranked by ROC-AUC (lr = Logistic Regression; gb = Gradient Boosting; xgb = XGBoost; mlp = Neural Network; rf =

Random Forest) For reference of the impact of ensembling, the best individual model was a hypertuned XGBoost (0.7701 ROC-AUC).

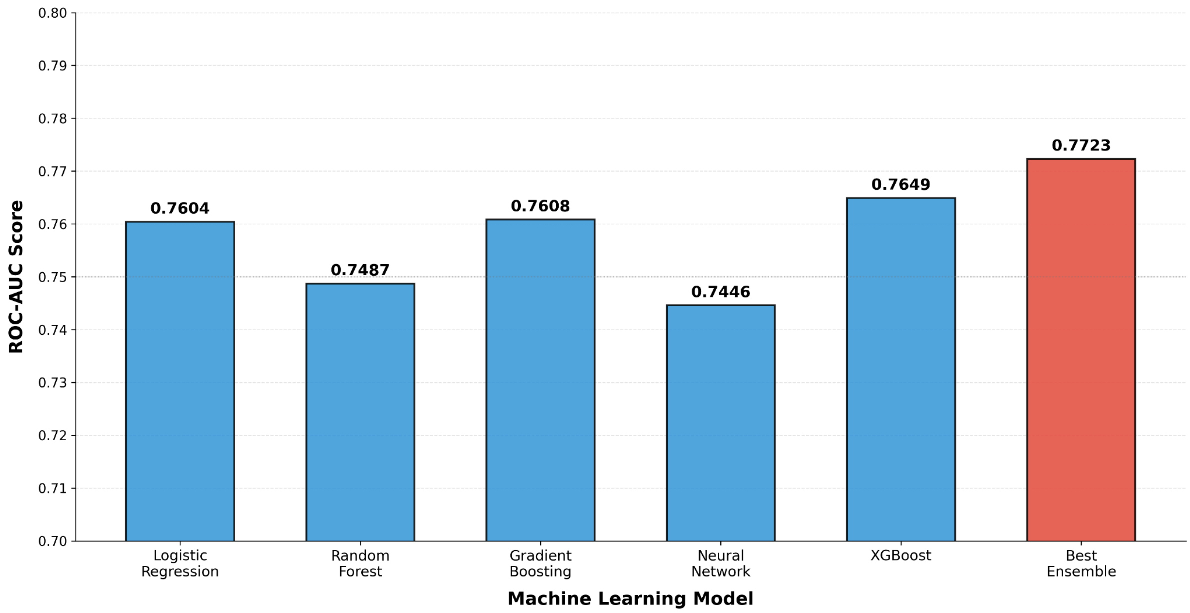

Figure 1 Model Performance Comparison ROC-AUC Scores**.** Best ROC-AUC scores for each of the five machine learning models and best ensemble (Stacking Classifier: Logistic Regression, Neural Network, and XGBoost). The Stacking ensemble achieved the highest performance (0.7723). Figure created using Python 3.12 with matplotlib and NumPy.

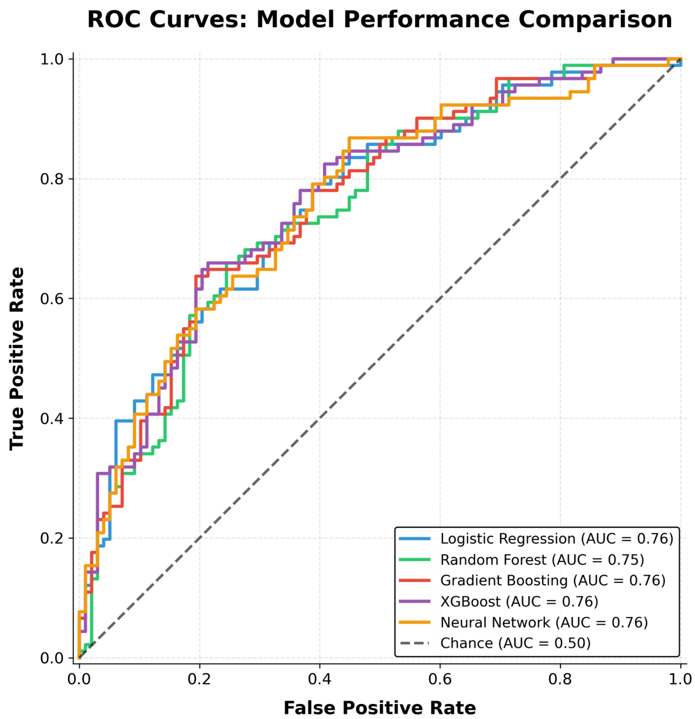

Figure 2 ROC curves showing each model's ability to distinguish between blooms and no-blooms across all possible decision thresholds. The further the curve bows toward the upper-left corner, the better the model performs. All models outperformed random guessing (diagonal dashed line, AUC = 0.50). Figure created using Python 3.12 with matplotlib and scikit-learn.

Decision Threshold and Final Model

The default prediction threshold for classification models is 0.5, meaning the model predicts a bloom whenever it estimates greater than 50% probability. However, in water quality monitoring, missing an actual bloom (false negative) is more dangerous than issuing an unnecessary warning (false positive). To reflect this, the threshold was lowered to 0.35, meaning the model predicts a bloom at a lower level of certainty to catch more actual blooms. This increases recall at the cost of more false alarms. Figure 5 shows the confusion matrix for the best performing model at the adjusted threshold of 0.35. The model correctly classified 52 true negatives and 77 true positives, while producing 46 false positives and 14 false negatives on the 189 lake test set, achieving an overall recall of 87.9%.

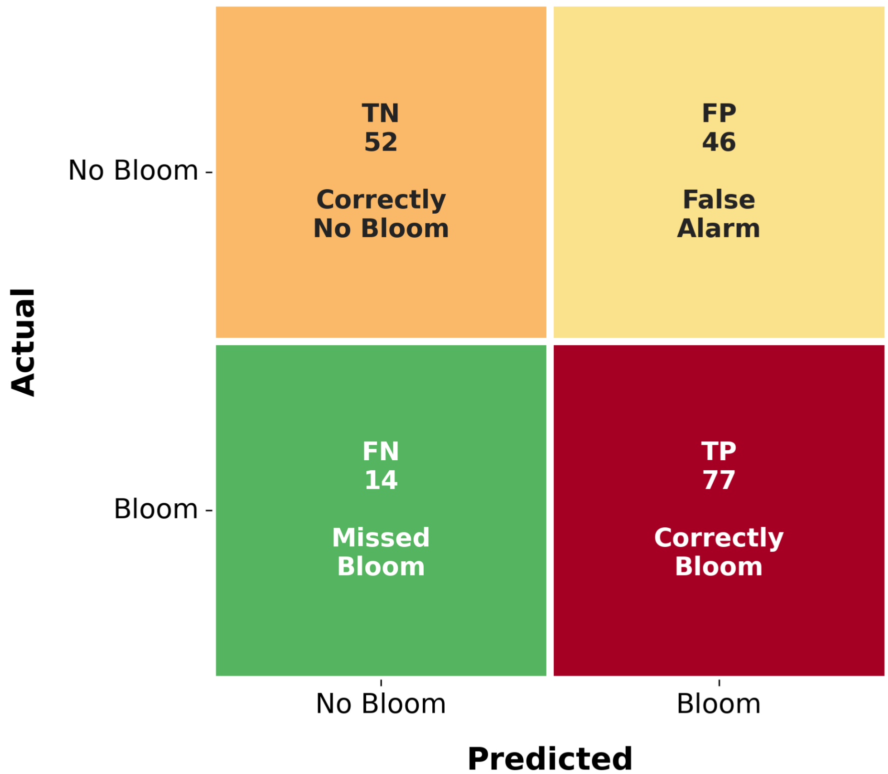

Figure 5 Confusion Matrix of Best Model**.** Confusion matrix showing how the best model (Stacking Classifier: Logistic Regression, Neural Network, and XGBoost classified the 189 test lakes. On the top left are the true negatives, top right are the false positives, bottom left are the false negatives, and bottom right are the true positives. The decision threshold was lowered to 0.35 to catch more actual blooms, accepting more false alarms as a safety trade-off. Figure created using Python 3.12 with seaborn and scikit-learn.

Discussion

The analysis evaluated the performance of the five algorithms trained on the four low-cost water quality sensors data through optimization stages of hyperparameter tuning, feature engineering, and ensembling (see Methodology). The overall best performance was from a Stacking Ensemble of the hyperparameter optimized Logistic Regression, XGBoost, and a Neural Network trained on the feature engineered dataset (ROC-AUC: 0.7723, 87.9% recall). The best individual model was the hyperparameter optimized XGBoost model trained on the feature engineered dataset (ROC-AUC: 0.7649). All 5 individual models exceeded baseline performance demonstrating validity of low-cost bloom prediction in majority cases. The results also revealed that while the best performance was an ensemble method, ensembling (Step 4) only showed marginal improvement from the best individual model (between +0.0035 and +0.0074 in ROC-AUC).

The results also revealed the effectiveness of other optimization techniques (hyperparameter tuning and feature engineering) in contributing to overall model performance. Overall, hyperparameter tuning improved the performance on average for the 5 models by around 8%. From that point, feature engineering added a marginal average improvement of around 2% in the ROC-AUC metric. In fact, the best individual model was a hyperparameter optimized XGBoost algorithm not trained on feature engineering. However, feature engineering's impact is not as limited as it may seem. It enabled the detection of important relationships in the data, which allowed for the performance increase. The engineered features proved essential for ensemble performance, as the stacking ensemble incorporating feature engineered models achieved the highest overall ROC-AUC (0.7723), demonstrating that the 2% average improvement was more significant than it might've seemed. Feature importance analysis revealed that turbidity (0.172) and pH (0.155) ranked as the strongest predictors, which aligns with expectations from previous research. The temperature-pH interaction (0.154) ranked third, confirming the impact of feature engineering on model training.

Overall, the top model performance was found to be promising. Achieving a ROC-AUC score of 0.7723 demonstrated the model's ability to distinguish between bloom and no-bloom conditions. This score was sufficient in this research to demonstrate real-world feasibility through stationary bot field testing. Additionally, the model was optimized for public safety by using a decision threshold of 0.35 compared to the standard 0.5. This adjustment increased recall to 87.9%, meaning the model successfully detects 88% of actual blooms in the test set. This came at the cost of increased false alarms, resulting in 46 false positives compared to 14 false negatives. This trade-off is appropriate for an early warning safety system where failing to detect an actual bloom poses greater risk than issuing an unnecessary advisory. The confusion matrix revealed 52 true negatives, 46 false positives, 14 false negatives, and 77 true positives, with the model correctly classifying 68% of test lakes overall. However, the 77% performance reflects limitations of using only these four low-cost water quality sensors. The model cannot capture critical bloom drivers including nutrient concentrations, light availability, and biological interactions that comprehensive EPA testing would provide.

Conclusion and Next Steps

This research successfully demonstrated the feasibility of inexpensive cyanobacterial bloom detection, supporting the hypothesis (ROC-AUC > 75%). The stacking ensemble model achieved 0.7723 ROC-AUC and 87.9% recall, successfully detecting the majority of blooms using only temperature, pH, conductivity, and turbidity measurements. Achieving a ROC-AUC of 77% with machine learning techniques like hyperparameter tuning and feature engineering proves the ability to use basic water quality metrics for effective monitoring. However, the 77% ROC-AUC represents the performance ceiling for just these four basic water quality sensors. Previous research has achieved scores upwards of 90% ROC-AUC, as other sensors are more directly relevant to cyanobacterial detection. This research's focus on affordability limited the accuracy of detections. Additionally, due to the lowered decision threshold for public safety concerns, the false alarm rate of the model is especially high. Despite this, the research demonstrated that detection can be both accessible and effective when used not as a standalone system but as a preliminary screening tool for under-resourced communities. The dramatic cost reduction, from thousands of dollars for comprehensive testing to less than $300, can make bloom detection accessible to communities lacking any monitoring at all. By demonstrating that machine learning can predict blooms with limited, affordable sensors, this work provides a foundation for equitable environmental monitoring in areas currently without any surveillance infrastructure. Thus, this research, despite its limitations, serves as a stepping stone for future work into accessible cyanobacterial threat detection.

The immediate next steps of this research would focus on improving detection accuracy. Given the focus on just four basic water quality sensors, limitations arose. I would look to overcome these limitations by investigating whether adding one or two more metrics to the training data (e.g., chlorophyll-a, fluorescence) can significantly benefit the performance of the machine learning algorithms while keeping the potential cost of the tool as low as possible. The goal would be to identify the minimum sensor set needed to achieve 85%+ ROC-AUC while maintaining affordability. Additionally, in order to validate the performance of the model deployed on the stationary floating platform, I would test the real-world performance across different lakes, geographic regions, and seasonal conditions to assess whether the model generalizes beyond the EPA National Lakes Assessment training data.

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